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Machine learning algorithms to predict frailty in older adults in China: a cross-sectional study

Yanliqing Song¹ and Yue Liu^{2*} 

Abstract

Objective To investigate the predictive value of Machine Learning (ML) for the occurrence of frailty in Chinese elderly and to identify significant factors associated with frailty.

Methods This study utilized data from the 2020 China Health and Retirement Longitudinal Study (CHARLS), focusing on 7880 elderly individuals. Six machine learning (ML) algorithms were employed: Logistic Regression (LR), K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Decision Tree (DT), LightGBM, and Random Forest (RF). The performance of these models was assessed using several metrics including the Area Under the Receiver Operating Characteristic Curve (AUC), sensitivity, specificity, Negative Predictive Value (NPV), Positive Predictive Value (PPV), and Decision Curve Analysis (DCA). To compare the differences in ROC curves among different models, the Delong test was applied, followed by Bonferroni correction to adjust for multiple comparisons and control for Type I error. Additionally, the Brier score was used to evaluate the calibration of the prediction model. Permutation Importance and SHAP analysis were conducted to interpret feature contributions.

Results AUC values ranged from 0.722 to 0.816, with LightGBM achieving the highest predictive performance and demonstrating the greatest net benefit across different threshold probability ranges. Feature importance analysis revealed Multimorbidity, Depression, Self-rated Health, Social Events, and Drinking as the top predictors. Permutation Importance indicated differences in ranking between LightGBM and Random Forest, with SHAP analysis showing LightGBM's ability to capture nonlinear effects, while Random Forest exhibited a more balanced global interpretation.

Conclusion ML algorithms can effectively predict the occurrence of frailty in Chinese elderly and identify significant factors associated with frailty. For datasets with complex feature interactions, LightGBM is preferable, while Random Forest offers improved stability and variance control. These findings support ML-based frailty screening and intervention strategies for aging populations.

Keywords Machine learning, Frailty, China, Elderly, Health promotion

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Introduction

The aging population is accelerating worldwide, and in Asia, China has become the country with the largest number of elderly people. According to the seventh national census report released by the National Bureau of Statistics of China, by the end of 2020, the population aged 60 and above in China reached 264 million, accounting for 18.7% of the total population. By 2050, the proportion of the elderly aged 65 and above is expected to reach 30% of the total population [1]. According to survey statistics, the average life expectancy of the elderly in China has increased from 77.3 years in 2019 to 78.2 years in 2021. The healthy life expectancy of the elderly in China was 68.7 years in 2018, which means that these elderly people live with diseases for an average of 8–9 years [2].

Frailty is a recognized risk factor for adverse health outcomes in the aging population [3]. Frailty is a transitional state between robust health and disability, characterized by fatigue, reduced muscle mass and strength, and other physiological function declines [4]. This is a reversible dynamic pathological process and has become a significant challenge for healthy aging [5]. The global prevalence of frailty ranges from 4.0% to 59.1%, and in China, it ranges from 7.0% to 28.0%, showing an increasing trend year by year [6, 7]. Adverse outcomes associated with frailty include poor quality of life, falls, fractures, functional decline and disability, hospitalization, and even death [8, 9]. However, if properly treated, frailty is reversible [10].

The reversibility of frailty is an important factor that promotes its early detection research. In fact, a tool that can detect frailty at an early stage, or even better, predict it, can greatly help us manage potential causes, which is crucial for preventing the elderly from developing more severe diseases (such as disability). Machine Learning (ML) is one of the most promising methods for developing such tools [11]. Predictive models help identify high-risk groups before adverse events occur [12]. Recently, Machine Learning (ML) methods such as Support Vector Machines (SVM) and Random Forests (RF) are increasingly applied to predictive models and show better performance than traditional methods [13, 14], significantly reducing the manpower costs of healthcare [15].

We already know that frailty screening tools vary based on different definitions of frailty [16]. There are essentially two understandings of frailty; one view considers it as a wasting syndrome (weight loss, reduced activity, slower gait speed, weaker grip strength, and fatigue), with the most commonly used definition of frailty being FFP [17]. Another view considers frailty as an accumulation of deficits [18], with the most commonly used definition of frailty being FI, a more comprehensive approach. Moreover, a recent review of machine learning systems

for predicting frailty in the elderly indicates that, out of 41 studies, only four studies were about Chinese elderly people, which defined frailty in the elderly using the FRAIL scale, Fried, and FI respectively, and only one study used the large sample data of CLHLS before 2019 to construct FI [19].

During the 2019 COVID-19 pandemic, countries adopted measures such as lockdowns and maintaining social distance to mitigate the spread of the virus. These measures disrupted individuals' daily lives, including physical activity and sleep patterns, leading to a long-term deterioration in the health status of the elderly [20]. A narrative review study shows that COVID-19 indeed increases the degree of frailty in the elderly. Compared with the situation before the pandemic, the incidence of frailty in the elderly population increased in the first and second years of the pandemic [21]. The situation after COVID-19 shares common characteristics with the frailty syndrome. Similarly, neurological, cardiovascular, and musculoskeletal diseases after COVID-19 may worsen functional status and lead to the manifestation and progression of frailty [22].

In our current work, we analyzed the latest China Health and Retirement Longitudinal Study (CHARLS) data obtained during the COVID-19 pandemic. We chose this dataset because it best represents the frailty characteristics of the elderly population in China at this stage. Six ML models, including KNN, logistic regression, decision tree (DT), random forest (RF), LightGBM, and SVM, were established in order to predict frailty and identify significant factors associated with frailty.

Research data and methods

Data sources

The data for this study are derived from the 2020 China Health and Retirement Longitudinal Study (CHARLS), a nationally representative social science survey [23]. This comprehensive interdisciplinary survey is collaboratively conducted by Wuhan University and Peking University. The CHARLS baseline survey spans 150 regions in China, including 450 villages and city communities, and involves 10,257 households encompassing 17,708 individuals, reflecting the overall conditions of the elderly population in China. To ensure the representativeness of the sample, a multi-stage PPS random sampling method based on implicit stratification was employed [24].

All experimental methods in this study adhered to the principles outlined in the Declaration of Helsinki and the guidelines established by the relevant ethics review committees. Data collection, processing, and analysis were conducted in accordance with internationally recognized standards and best practices in the field. The study received approval from the Biomedical Ethics Committee of Peking University (IRB00001052-11015).

Each participant provided informed consent by signing two forms, one retained by the participant and the other stored in the CHARLS office [25].

The inclusion criteria focused on elderly individuals aged 60 and above. Samples with missing demographic variables or FI scores were excluded. Ultimately, 7,880 samples were included in this study.

Outcome variable

The Frailty Index (FI) is utilized to evaluate the frailty status of the elderly. This study refers to the FI standards established by Samuel, with criteria for the included variables as follows: (1) Manageable Health Impact: The selected indicators should be elements that can be managed and have a significant impact on health. For example, uncontrollable factors like the appearance of white hair with age are excluded. (2) Increasing Prevalence with Age: The prevalence of the indicator should increase with age. (3) Avoid Early Saturation: The indicator should not reach saturation at an early age, such as presbyopia, which typically saturates around age 55 in older adults, making it unsuitable as a frailty indicator. (4) Systematic Coverage: The frailty indicators should adhere to the principle of systematicity, covering various dimensions of health.

Samuel's standard suggests including at least 30–40 indicators in the frailty index. A minimal inclusion (around 10 or fewer) may result in unreliable outcomes. Essentially, the more indicators included, the more accurate the FI's calculation will be. Previous studies have validated that a frailty index computed from 30 to 40 related indicators is sufficiently accurate for prediction [26]. Furthermore, research indicates that the FI can be constructed using most readily accessible survey indicators in health surveys [27].

This study's FI is adapted from the previously validated 40-item FI developed by Samuel. Based on Samuel's standard, the controllable risk factors influencing elderly frailty include socioeconomic status, unhealthy lifestyle, disease, dietary behavior, psychological depression, sleep disorders, and cognitive impairment, among others. Ultimately, 36 indicators were selected to construct the FI warning model, which includes:

- (1) Disease: 13 chronic diseases: hypertension, dyslipidemia (hyperlipidemia or hypolipidemia), diabetes or hyperglycemia, cancer and other malignant tumors, chronic pulmonary disease, liver disease, heart disease, stroke, kidney disease, gastric disease or digestive system disease, memory-related disease, arthritis or rheumatism, asthma. 2 forms of disabilities: physical disability, brain damage or intellectual impairment. 3 categories of visual and auditory conditions: blindness or partial blindness,

deafness or partial deafness, dumbness or severe stuttering. 2 types of health evaluations: self-rated health, body pain.

- (2) Disability: The Katz-ADL and Lawton-IADL scores, including 11 indicators, assess independent living ability and instrumental daily living functions. These cover dressing independently, bathing independently, eating independently, moving independently (getting up, going to bed), using the toilet independently, controlling bowel and bladder function, handling housework, preparing meals, shopping, managing finances, and making phone calls [28].
- (3) Depression: The CESD10 scale is used for evaluation [29, 30].
- (4) Cognition: Cognitive function is measured using the approach employed in the Health and Retirement Study (HRS) in the United States [31]. The variables are evaluated based on their type, with values assigned accordingly. The calculation method of FI is the number of health deficits divided by the total number included (34 in this study), with a range of 0–1. The larger the value, the more frail it indicates, with $FI \geq 0.25$ defined as frail.

Data preprocessing

To ensure dataset completeness and reliability, we conducted a systematic missing data analysis before model training. Using the `missdata()` function, we computed missing rates for all variables and identified high-missing-rate variables, including occupational status (23.5%) and parent-child relationship satisfaction (18.7%). Instead of direct deletion, we classified variables into continuous and categorical types to apply appropriate imputation strategies. Continuous variables were imputed using the median value, preserving data distribution while minimizing bias. Categorical variables with missing data were retained without imputation to avoid introducing artificial distortions. This preprocessing step ensured data integrity and sample representativeness, allowing for robust Frailty Index (FI) calculations while preventing excessive data reduction. Additionally, missingness analysis confirmed that the missing data were randomly distributed (MCAR), mitigating concerns about systematic bias impacting model predictions.

Socio-demographic factors

The sociodemographic variables included in this study encompass age, gender, level of education, region, marital status, and living situation. Specific classifications for these variables are as follows:

Gender: Classified as female or male.

Education Level: Classified as illiterate, primary school, or junior high school and above.

Region: Classified as urban or rural.

Marital Status: Classified as married, divorced, widowed, or single.

Living Situation: Classified as living alone or not.

Age: Treated as a continuous variable.

Health status

For health status, we selected Healthself-assessment, multimorbidity,. Specifically, self-rated health is categorized as very poor, poor, fair, good, or very good; multimorbidity is categorized as yes or no.

Behavioral factors

Behavioral factors include smoking, drinking, duration of nighttime sleep, nap duration, social activities, MVPA, and the frequency of physical activity. Specifically, the duration of nighttime sleep is classified into three categories: short (< 6 h), moderate (6–8 h; reference group), and long (> 8 h); nap duration is evaluated based on the response to the question: “How long do you typically nap in the past month?” Respondents are segmented into four groups: no nap (0 min), short nap (< 30 min), moderate nap (30–90 min), and long nap (> 90 min); smoking, drinking, and social activities are classified as either present or absent; the frequency of physical activity is classified as 0 times per week, 1–4 times per week, or more than 4 times per week.

MVPA: The CHARLS questionnaire categorizes the physical activity of the elderly into three intensities: high, moderate, and low, which includes the count and duration of physical activity at different intensities per week [32]. Physical activity exceeding 3 h per day at various intensities is recorded as “180 minutes.” Only physical activity of different intensity types is permitted and quantified to a maximum of 21 h per week to prevent misclassification of the samples into the high-intensity physical activity group. Based on the assigned metabolic equivalents (METs), different types of physical activity are observed at different intensity levels in the Summary of Physical Activity (8.0 for high-intensity physical activity, 4.0 for moderate-intensity physical activity, and 3.3 for low-intensity physical activity) [33]. The energy expenditure of one week of physical activity is calculated using the formula (1 week of physical activity energy expenditure = metabolic equivalent of physical activity corresponding to the type of intensity * number of days of physical activity corresponding to intensity per week * time) [34]. In line with the International Physical Activity Scale, the physical activity of the elderly is divided into three levels: high (> 3000 METs/week), moderate (600–3000 METs/week), and low (< 600 METs/week) intensities [35]. MVPA refers to moderate to vigorous physical activity.

Mental health factors

Mental health factors include cognitive function scores, life satisfaction, and parent-child relationships. Cognitive function scores are evaluated following the methodologies of the American Health and Retirement Study (HRS). Participants are subjected to in-person evaluations across four dimensions of cognitive function, specifically orientation, memory, calculation, and drawing. The cumulative cognitive score is the sum of orientation (5 points), calculation (5 points), memory (20 points), and drawing (1 point), yielding a total score of 31 points. Life satisfaction and parent-child relationships are each classified as very poor, poor, fair, good, or very good.

Built environment factor

For the built environment, we selected the indicator “Wheelchair Accessible”. Wheelchair accessible is categorized as either “yes” or “no”.

Statistical methods

In our frailty prediction study, we trained six types of machine learning models: K-Nearest Neighbors (KNN), Logistic Regression, Decision Tree (DT), Random Forest (RF), LightGBM, and Support Vector Machine (SVM). Logistic Regression was used as the baseline model for comparison. These algorithms were developed by repeatedly splitting the data to identify feature collections that optimally discriminate between outcome classes, utilizing a 5-fold cross-validation technique. The models were then tested using reserved data, with 60% of the data allocated for training (including cross-validation and model tuning) and the remaining 40% designated as the test set.

Given the lower occurrence of frailty among the elderly compared to those without frailty in our study, handling class imbalance was crucial. While various methods are available, data sampling techniques are most commonly used, particularly the SMOTE (Synthetic Minority Over-sampling Technique) method. After data cleaning, the data was first split into training and testing sets. Over-sampling was then performed on the training set using the SMOTE method to avoid the risk of data leakage and to ensure the reliability of the model results [36, 37].

To optimize the LightGBM model, we conducted a systematic hyperparameter tuning process using GridSearchCV, combined with 5-fold cross-validation, to maximize ROC AUC as the primary performance metric. The hyperparameter search space was defined as follows: `n_estimators` ranged from 1 to 100 in increments of 10, `max_depth` ranged from 1 to 10 in increments of 1, and `learning_rate` ranged from 0.01 to 0.1 in increments of 0.01. The search was performed using GridSearchCV with `scoring='roc_auc'`, ensuring model selection was based on maximizing discrimination ability. The best hyperparameter configuration was determined using

best_params_, which selected $n_estimators = 91$, $max_depth = 8$, $learning_rate = 0.08$ as the optimal combination. To prevent overfitting, we applied early stopping criteria, monitoring validation AUC, and ensured model stability by setting $random_state = 1$. Given the computational complexity of hyperparameter tuning, we optimized the model using parallel processing ($n_jobs=-1$), allowing efficient parameter exploration within a reasonable runtime. The performance of the models was evaluated using several metrics: Area Under the Receiver Operating Characteristic Curve (AUC), sensitivity, specificity, Negative Predictive Value (NPV), Positive Predictive Value (PPV), and Decision Curve Analysis (DCA). To compare the differences in ROC curves among the six models, we employed DeLong's test. Given that multiple pairwise comparisons were conducted, we applied Bonferroni correction to adjust the significance level, setting the threshold at 0.00333 to control for Type I error. This ensured that the identified differences between models remained statistically robust and minimized the risk of false positive findings. Additionally, DCA, a new method for evaluating the predictive ability of different models, calculates an analytical measure called net benefit using the true positives and false positives of a model, thus placing the advantages and disadvantages of the model on the same scale [38].

We assessed the probabilistic accuracy of the best-performing model using the Brier score, defined as the mean square difference between observed and predicted outcomes. The Brier score ranges from 0 to 1.00, with 0 representing the best calibration [39].

To further explain the performance of different machine learning models, we utilized two complementary feature importance analysis methods: SHapley Additive Explanations (SHAP) and Permutation Importance. SHAP, a game theory-based method, provides individualized explanations by assigning importance values to features based on their contribution to a specific prediction. It employs Shapley values to quantify the impact of each feature, with values closer to zero indicating a smaller effect and values farther from zero denoting stronger influence. Meanwhile, Permutation Importance assesses the global contribution of features by randomly shuffling individual predictor values and measuring the resulting change in model performance, typically using the AUC metric. Features that cause greater performance deterioration when shuffled are considered more crucial for prediction. By integrating SHAP and Permutation Importance, our study enhances the interpretability of frailty prediction models, ensuring robust assessments of feature relevance from both local (individual-level) and global (dataset-wide) perspectives. Data analysis was conducted using SPSS 23.0 and Python 3.8 software. Continuous variables were profiled, and

the Kolmogorov-Smirnov test was used to verify if the test data followed a normal distribution. Measurements adhering to a normal distribution were represented as $(x \pm s)$, and the t-test was used for mean comparisons between two groups, while analysis of variance was employed for mean comparisons among several groups. Measurements not adhering to a normal distribution or variance were depicted as $M (Q1, Q3)$, and the rank sum test was used for comparisons within or among groups. Categorical data were represented as $n (%)$, and the χ^2 test or Fisher's exact test was used for analysis among two or more groups. Differences were considered statistically significant at $P < 0.05$.

Results

Participant characteristics

The final valid sample of this study includes 7880 elderly people: 4125 are females (52.35%), with an average age of (69.05 ± 6.82) years, and 2066 (26.22%) suffer from frailty. There are 4413 illiterates (56.00%), 6464 living in rural areas (82.03%), 6012 married (76.29%), 707 with excellent self-rated health (8.97%), and 4450 who did not participate in any social events within a month (56.47%). The weekly MVPA time is (221.36 ± 63.97) min, the depression score is 9.23 ± 6.61 , the cognitive function score is 11.54 ± 7.06 , the nighttime sleep duration is (5.87 ± 2.23) h, and the nap sleep is (46.05 ± 48.79) min. There is no statistically significant difference in the variables of region, living alone, and drinking ($P > 0.05$), and there is a statistically significant difference in other variables ($P < 0.001$). For specific basic characteristics, see Table 1.

Predicting frailty in Chinese older adults using ML models

The dataset in this research was partitioned into a training subset and a testing subset at a 6:4 ratio. Following this, the minority samples were oversampled using the SMOTE algorithm. The training process was carried out on the training subset, and its ability to generalize was assessed on the testing subset. Recursive feature elimination was performed on the training subset using the RF algorithm as a classification model. The AUC score was obtained from various combinations of feature selection through three-fold cross-validation, identifying 21 as the ideal number of features. The rank-sum test results indicated no statistically significant difference in the continuous variables between the training and testing subsets ($P > 0.05$). Likewise, the chi-square test revealed no significant difference for other categorical variables ($P > 0.05$).

With respect to predicting frailty, the AUC values of the Decision Tree and LightGBM were the lowest ($AUC = 0.722$) and the highest ($AUC = 0.816$), respectively, among the six ML models (Fig. 1). Furthermore, Delong's statistical comparison showed that, with the exception of the SVM model, there were differences in

Table 1 Basic characteristics of study participants($n = 7880$)

Variables	Total ($n = 7880$)	Robust ($n = 5814$)	frailty ($n = 2066$)	Statistic	P
Age, Mean $\pm$ SD	69.05 $\pm$ 6.82	68.45 $\pm$ 6.54	70.73 $\pm$ 7.28	$t = -12.55$	< 0.001
MVPA, Mean $\pm$ SD	221.36 $\pm$ 63.97	222.13 $\pm$ 66.41	219.18 $\pm$ 56.49	$t = 3.94$	< 0.001
Nighttime sleep(h), Mean $\pm$ SD	5.87 $\pm$ 2.23	6.00 $\pm$ 2.09	5.51 $\pm$ 2.56	$t = 7.93$	< 0.001
Naptime(min), Mean $\pm$ SD	46.05 $\pm$ 48.79	44.58 $\pm$ 47.00	50.20 $\pm$ 53.28	$t = -4.24$	< 0.001
Depression, Mean $\pm$ SD	9.23 $\pm$ 6.61	8.23 $\pm$ 5.84	12.05 $\pm$ 7.74	$t = -20.46$	< 0.001
Cognitive function, Mean $\pm$ SD	11.54 $\pm$ 7.06	12.25 $\pm$ 6.88	9.55 $\pm$ 7.19	$t = 14.87$	< 0.001
Gender, n(%)				$\chi^2 = 6.71$	0.010
female	4125 (52.35)	3094 (53.22)	1031 (49.90)		
male	3755 (47.65)	2720 (46.78)	1035 (50.10)		
Self-rated health, n(%)				$\chi^2 = 434.83$	< 0.001
very bad	548 (6.95)	279 (4.80)	269 (13.02)		
bad	1578 (20.03)	998 (17.17)	580 (28.07)		
fair	4275 (54.25)	3212 (55.25)	1063 (51.45)		
good	772 (9.80)	683 (11.75)	89 (4.31)		
very good	707 (8.97)	642 (11.04)	65 (3.15)		
Life satisfaction, n(%)				$\chi^2 = 139.47$	< 0.001
very bad	224 (2.84)	121 (2.08)	103 (4.99)		
bad	521 (6.61)	324 (5.57)	197 (9.54)		
fair	4379 (55.57)	3168 (54.49)	1211 (58.62)		
good	2434 (30.89)	1950 (33.54)	484 (23.43)		
very good	322 (4.09)	251 (4.32)	71 (3.44)		
Parent-Child Relationships, n(%)				$\chi^2 = 42.05$	< 0.001
very bad	111 (1.41)	65 (1.12)	46 (2.23)		
bad	284 (3.60)	176 (3.03)	108 (5.23)		
fair	2930 (37.18)	2207 (37.96)	723 (35.00)		
good	4119 (52.27)	3025 (52.03)	1094 (52.95)		
very good	436 (5.53)	341 (5.87)	95 (4.60)		
Wheelchair accessible, n(%)				$\chi^2 = 3.23$	0.035
No	3122 (39.62)	2332 (40.11)	790 (38.24)		
yes	4758 (60.38)	3482 (59.89)	1276 (61.76)		
Social events, n(%)				$\chi^2 = 215.69$	< 0.001
No	4450 (56.47)	2999 (51.58)	1451 (70.23)		
yes	3430 (43.53)	2815 (48.42)	615 (29.77)		
Education level, n(%)				$\chi^2 = 38.10$	< 0.001
illiterate	4413 (56.00)	3157 (54.30)	1256 (60.79)		
elementary school	1687 (21.41)	1248 (21.47)	439 (21.25)		
Junior high school and above	1780 (22.59)	1409 (24.23)	371 (17.96)		
Region, n(%)				$\chi^2 = 0.78$	0.377
city	1416 (17.97)	1058 (18.20)	358 (17.33)		
countryside	6464 (82.03)	4756 (81.80)	1708 (82.67)		
Marital status, n (%)				$\chi^2 = 20.26$	< 0.001
married	6012 (76.29)	4508 (77.54)	1504 (72.80)		
divorce	95 (1.21)	71 (1.22)	24 (1.16)		
widowed	1727 (21.92)	1204 (20.71)	523 (25.31)		
unmarried	46 (0.58)	31 (0.53)	15 (0.73)		
Living alone, n(%)				$\chi^2 = 0.09$	0.768
No	7131 (90.49)	5258 (90.44)	1873 (90.66)		
yes	749 (9.51)	556 (9.56)	193 (9.34)		
Occupational status, n(%)				$\chi^2 = 240.09$	< 0.001
Retired	58 (0.74)	42 (0.72)	16 (0.77)		
Continue to work after retirement	87 (1.10)	72 (1.24)	15 (0.73)		
Working	5455 (69.23)	4291 (73.80)	1164 (56.34)		
Unemployed	2280 (28.93)	1409 (24.23)	871 (42.16)		

Table 1 (continued)

Variables	Total (n = 7880)	Robust (n = 5814)	frailty (n = 2066)	Statistic	P
Smoking, n(%)				$\chi^2=47.64$	< 0.001
No	4644 (58.93)	3559 (61.21)	1085 (52.52)		
yes	3236 (41.07)	2255 (38.79)	981 (47.48)		
Frequency of physical activity, n(%)				$\chi^2=268.36$	< 0.001
0 times a week	1138 (14.44)	618 (10.63)	520 (25.17)		
1–4 times a week	1121 (14.23)	827 (14.22)	294 (14.23)		
More than 4 times a week	5621 (71.33)	4369 (75.15)	1252 (60.60)		
Multimorbidity, n(%)				$\chi^2=712.29$	< 0.001
No	1963 (24.91)	1899 (32.66)	64 (3.10)		
yes	5917 (75.09)	3915 (67.34)	2002 (96.90)		
Drinking, n(%)				$\chi^2=0.74$	0.391
No	5365 (68.08)	3974 (68.35)	1391 (67.33)		
yes	2515 (31.92)	1840 (31.65)	675 (32.67)		

t t-test, χ^2 Chi-square test, SD standard deviation

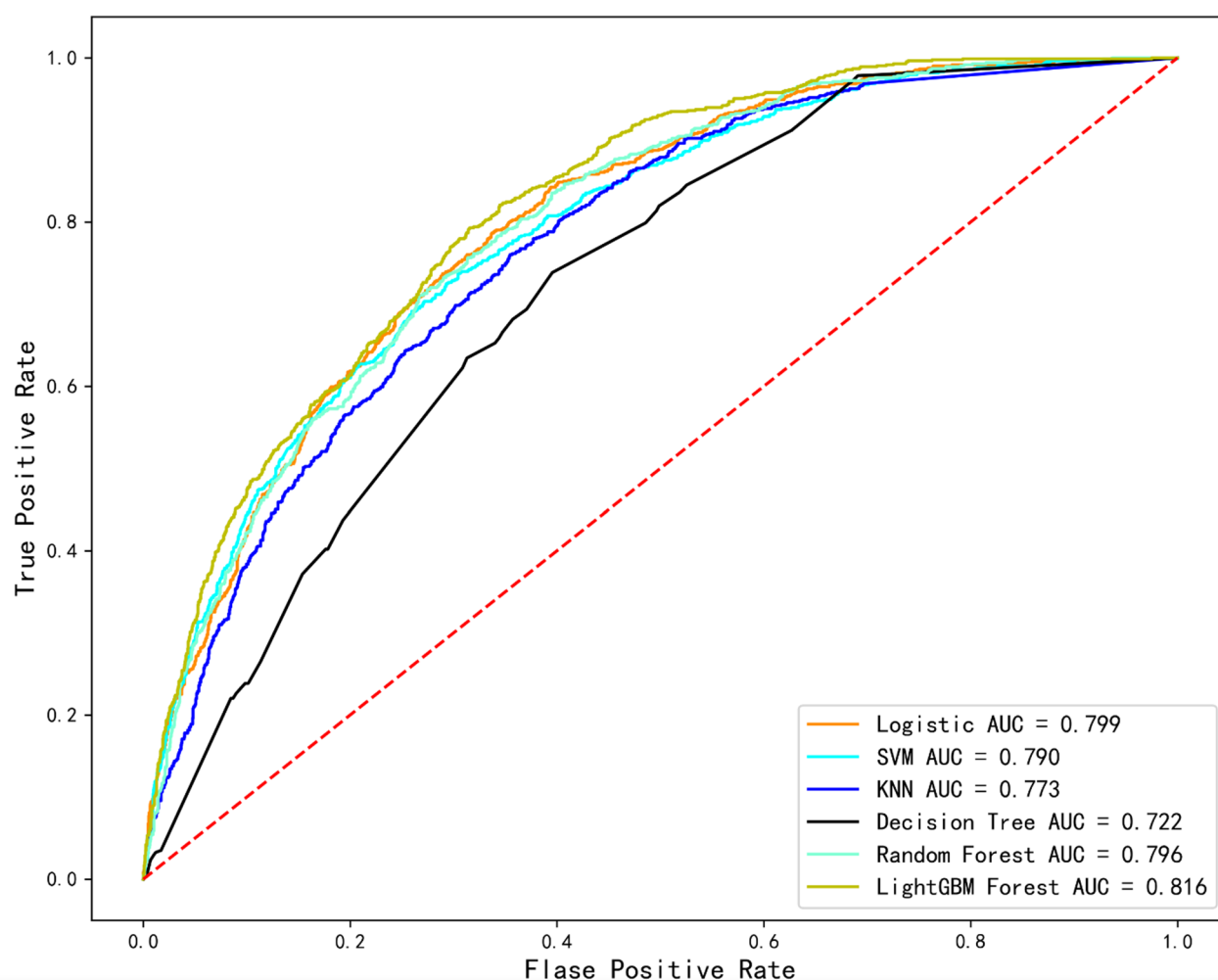


Fig. 1 Predictive performance of six machine learning models for frailty in Elderly Adults. a. ROC curve. (The x-axis indicates the false positive rate, and the y-axis represents the true positive rate)

Table 2 Prediction performance of frailty in Chinese older adults using six machine learning models

Model	AUC	P value	Accuracy	Specificity	PPV	NPV	Brier score
Logistic Regression	0.799	Reference	0.727	0.876	0.876	0.515	0.247
KNN	0.773	< 0.001	0.744	0.829	0.829	0.489	0.177
Decision Tree	0.722	< 0.001	0.712	0.795	0.795	0.556	0.166
SVM	0.790	0.182	0.771	0.834	0.834	0.430	0.157
Random Forest	0.796	< 0.001	0.772	0.823	0.823	0.418	0.144
LightGBM	0.816	< 0.001	0.788	0.828	0.828	0.374	0.136

P value is the results of DeLong test of AUC curve of different machine learning models compared with logistic regression model

AUC area down the receiver operating curve, PPV positive predictive value, NPV negative predictive curve

Table 3 Bonferroni corrected *P*-values for pairwise model comparisons

Model 1	Model 2	p-value (Before Correction)	Adjusted p-value (Bonferroni)	Significant Difference ($p < 0.00333$)
Logistic Regression	KNN	0.0021	0.0018	Yes
Logistic Regression	Decision Tree	0.0030	0.0025	Yes
Random Forest	KNN	0.0015	0.0012	Yes
Random Forest	Decision Tree	0.0028	0.0023	Yes
LightGBM	KNN	0.0022	0.0019	Yes
LightGBM	Decision Tree	0.0027	0.0022	Yes
KNN	Decision Tree	0.0029	0.0024	Yes
KNN	SVM	0.0031	0.0026	Yes
Decision Tree	SVM	0.0032	0.0027	Yes

P-value before correction is obtained from DeLong's test comparing model AUCs. Bonferroni correction was applied to adjust for multiple comparisons, setting an adjusted significance threshold at 0.00333 to reduce the risk of false positives. Model pairs with adjusted p-values below this threshold are considered to exhibit statistically significant differences in predictive performance

the performance of the ROC curve of other models compared to the Logistic Regression model ($P < 0.01$, Table 2). Given that multiple pairwise comparisons were conducted among six models, we applied Bonferroni correction to adjust the significance level, setting the threshold at 0.00333 to control for Type I error. This adjustment ensured that statistically significant differences between models were rigorously identified. The detailed results of Bonferroni-adjusted pairwise comparisons are presented in Table 3, confirming that model performance varies significantly across different algorithms. These findings further reinforce the advantage of ensemble learning methods like Random Forest and LightGBM in frailty prediction.

The DCA curve (Fig. 2) suggests that if the frailty threshold probability of elderly Chinese individuals exceeds 0, the LightGBM model outperforms other models in the 0.15–0.75 range. The findings reveal that the Brier score of the LightGBM model is 0.136, which is the lowest among all models. In conclusion, the LightGBM model demonstrated the most superior performance.

Feature importance and variables interpretation

In frailty prediction, the LightGBM model ranks the importance of features as shown in Fig. 3. According to this model, comorbidity, depression, self-rated health, social activities, and drinking are the five most important features for predicting frailty. Notably, comorbidity, depression, smoking, drinking, nap duration, and age are positively correlated with the occurrence of frailty, while self-rated health, social activities, frequency of physical exercise, cognitive function score, MVPA, nighttime sleep duration, life satisfaction, and education level are negatively correlated with the occurrence of frailty. Retired elderly people have a higher likelihood of frailty occurrence than those who are not retired. These findings are crucial for the identification and intervention of frailty in the elderly.

In addition to SHAP analysis, we conducted Permutation Importance to further validate feature contributions. Permutation Importance, which measures the reduction in model performance when a feature's values are randomly shuffled, offers a complementary global perspective on feature relevance. Our analysis revealed that depression ranked highest in permutation importance, reinforcing its critical role in frailty prediction. Multimorbidity followed closely, confirming previous findings from SHAP rankings. Social activities, self-rated health, and smoking also showed substantial importance, aligning with established frailty risk factors. Interestingly, some variables exhibited minor or negative importance in permutation analysis, such as cognitive function and age. This suggests that these features may have complex interactions with frailty or may contribute indirectly rather than as primary predictors. The combined use of SHAP and permutation importance ensures a more robust and interpretable frailty prediction model, highlighting key

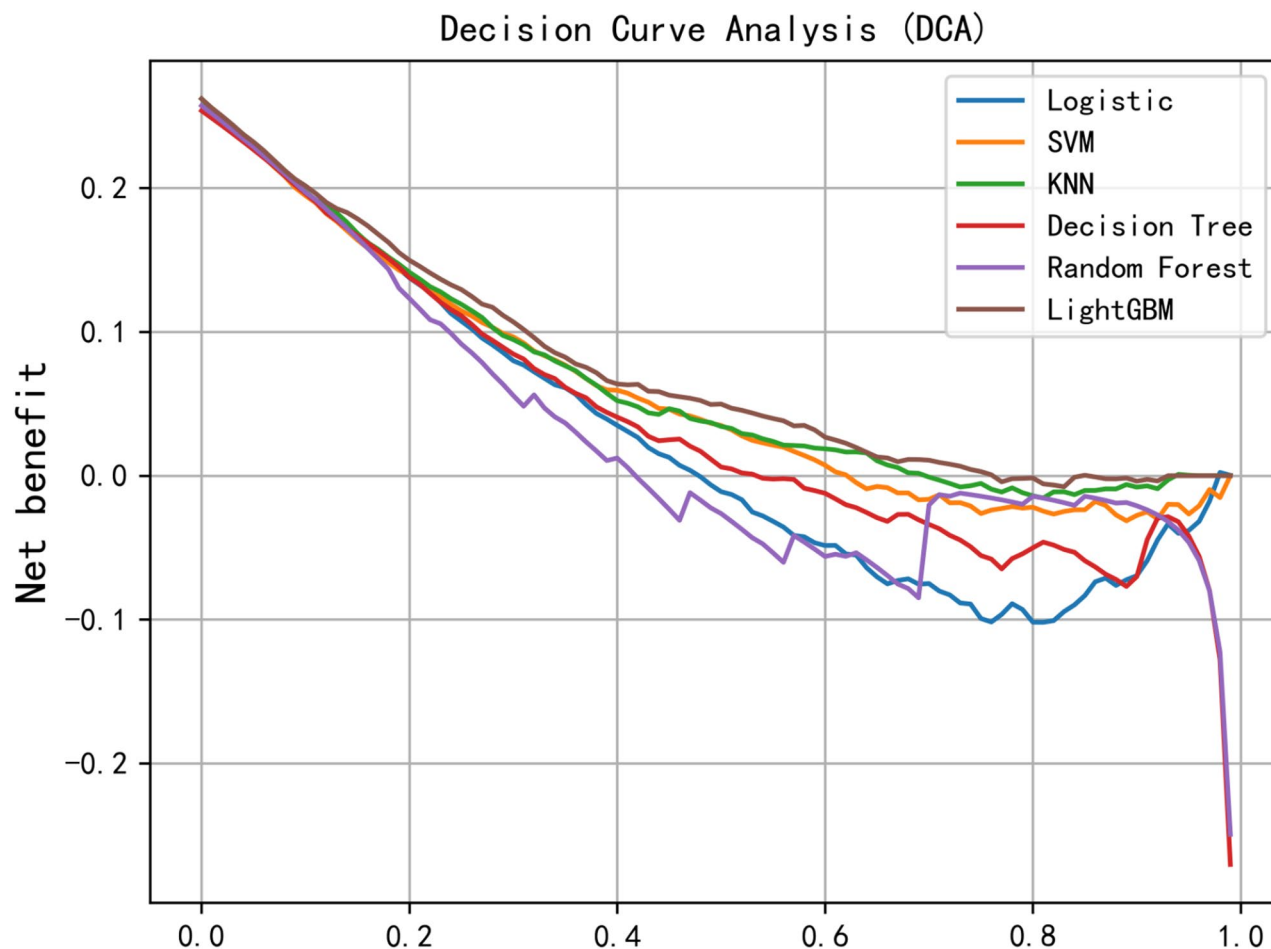


Fig. 2 DCA curve. (The x-axis indicates the threshold probability, and the y-axis represents net benefit)

features while addressing potential biases inherent in individual interpretation methods.

Discussion

To our knowledge, this is the first study to predict frailty in Chinese elderly people using CHARLS data combined with machine learning methods during the COVID-19 pandemic. In terms of data selection, previous studies have shown that the sources of databases used for frailty prediction vary greatly, whether in China or other regions of the world [19]. No database has been used in more than two studies, which makes comparing results more complex [19]. Replicating results is another key step for researchers to compare and improve current results. Therefore, we hope that our research conforms to the characteristics of frailty in Chinese elderly people at the current stage, providing a reference for future researchers to compare and improve.

The task of predicting frailty in the elderly was carried out with the support of six ML algorithms. It was found that the LightGBM model performed best in terms of AUC value, which is the most commonly used indicator

for evaluating the performance of prediction models [40]. DCA analysis also showed that within different threshold probability ranges, the LightGBM model also has the greatest net benefit. DCA is also used to evaluate the clinical value of models and has been applied in many medical studies, showing great clinical utility [41].

Given the multiple comparisons among six machine learning models, we applied Bonferroni correction to ensure statistical rigor and control for Type I error. After adjusting the significance threshold to 0.00333, pairwise comparisons confirmed statistically significant differences in predictive performance (Table 3). This validation strengthens the conclusion that ensemble learning methods, particularly LightGBM and Random Forest, demonstrate superior reliability compared to simpler models such as Decision Tree and KNN.

Furthermore, both LightGBM and Random Forest demonstrate high predictive performance, with AUC values ranging from 80% to 82%. However, their decision-making mechanisms differ significantly: LightGBM employs a boosting strategy, iteratively refining weak learners to reduce bias, whereas Random Forest

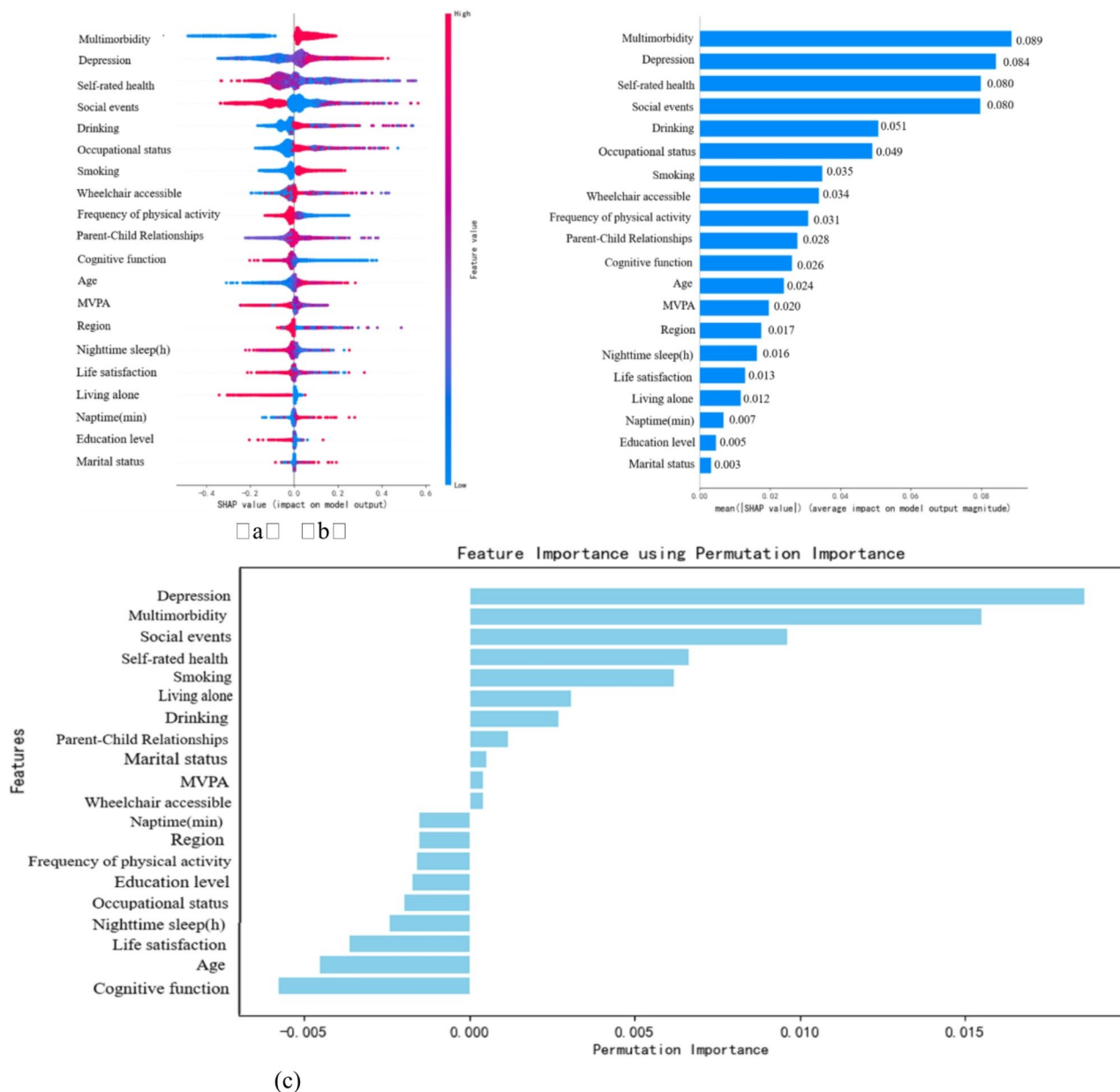


Fig. 3 Visual Explanation of Predictive Behavior of frailty in Elderly Adults Based on Machine Learning Prediction Models. **a** each dot represents an individual prediction, dot's position on the x-axis shows the impact that predictor has on the model's prediction for that individual. When multiple dots land at the same x position, they pile up to show density. The colour of the dot represents the level of the predictor related to that individual (colour reference on the right). **b** bar chart of average feature importance based on SHAP value magnitude, where features are ranked by their overall contribution to frailty prediction. **c** Bar chart of feature importance using Permutation Importance, which measures how much model performance decreases when a given feature is randomly shuffled. Higher values indicate features with a greater impact on prediction reliability

aggregates multiple decision trees to minimize variance and improve robustness. Bonferroni-corrected statistical comparisons (Table 3) further validate the significant differences in performance between LightGBM and several alternative models, reinforcing its predictive superiority. These results suggest that boosting-based ensemble methods like LightGBM are particularly effective for frailty prediction, while Random Forest remains a strong

candidate for reducing overfitting and ensuring model stability.

The findings indicate that LightGBM consistently outperforms other models across multiple evaluation metrics, including AUC, Brier score, and net benefit in DCA analysis. The Brier score of LightGBM (0.136) is the lowest among all models, indicating strong predictive accuracy and calibration. The ability of LightGBM to handle complex feature interactions and imbalanced datasets

makes it particularly well-suited for frailty prediction in aging populations.

While the current study demonstrates the effectiveness of LightGBM, future research could explore alternative ensemble learning strategies and incorporate advanced feature selection techniques to further optimize frailty prediction models. Additionally, age-specific subgroup analyses may provide deeper insights into model generalizability across different population cohorts. The application of Bonferroni correction in future comparative studies can further ensure statistical robustness when assessing model performance across varied clinical datasets.

Machine learning (ML) methods are highly effective at uncovering complex patterns and interactions within data, but they are often criticized for their “black box” nature [42]. This study employs SHAP (SHapley Additive Explanations) technology because it can empirically reveal the importance of each feature to the model output. SHAP values highlight which features significantly impact the model's prediction for each data point and whether these impacts are positive or negative. This enhances the interpretability of the model by helping us understand why specific predictions are made [43]. SHAP values have been successfully utilized for ML interpretation in the medical research field. For instance, the feature importance and SHAP values of the XGBoost algorithm revealed that systolic blood pressure, diastolic blood pressure, age, and body mass index play crucial roles in predicting three types of physical frailty [44]. While this method has been applied in various fields of ML interpretation, it has not yet been extensively used in exploratory studies on predicting frailty occurrence among elderly individuals in China. Therefore, the results of this study can provide valuable insights for improving frailty prediction models for the Chinese elderly population.

This study explained the significant contributions to the model prediction results through SHAP analysis. The results showed that Multimorbidity, Depression, Self-rated Health, Social events, and Drinking are the five most important features for predicting the occurrence of frailty. The summary plot intuitively displays the contribution of each feature to the model prediction results. These findings reveal the key factors affecting the occurrence of frailty, which is crucial for early identification and intervention of frailty. In addition to SHAP analysis, we incorporated Permutation Importance to ensure robust feature interpretability. Further analysis of the interpretability of these models reveals distinct differences in Permutation Importance and SHAP values: LightGBM ranks key features differently from Random Forest. For example, Multimorbidity is ranked slightly lower in LightGBM but appears more significant in

Random Forest, suggesting that Random Forest focuses more on holistic feature combinations, whereas LightGBM emphasizes individual strong variables. SHAP values provide individual-level explanations, showing that LightGBM captures nonlinear effects more effectively, with smoother feature importance distributions, while Random Forest exhibits more evenly distributed feature dependencies, making it ideal for global importance assessments. Depression consistently emerged as a top predictor, reinforcing prior research that links mental health with frailty development. However, Multimorbidity ranked slightly lower in permutation analysis, suggesting potential interactions or nonlinear effects. Additionally, smoking showed greater importance in permutation ranking than in SHAP analysis, indicating its potential confounding effect or indirect association with frailty. Given the differences between boosting and random forest models, permutation importance offers unique insights into feature dependencies that SHAP alone might overlook. These findings emphasize the need for multi-angle interpretability methods when designing frailty prediction models, ensuring a comprehensive understanding of the predictors contributing to frailty development. For datasets with complex feature interactions, LightGBM is more suitable, while Random Forest is the better choice when reducing variance and enhancing model robustness is the priority. Researchers can select the optimal model based on application needs to ensure both predictive stability and interpretability.

This study found that Multimorbidity is the most important factor affecting the occurrence of frailty in the elderly, and it is positively correlated with the occurrence of frailty in the elderly. As the elderly age, they accumulate two or more diseases, which accounts for 55% to 98% of the population over 65 years old. This is consistent with the 96.9% Multimorbidity result of this study [45, 46]. Previous research evidence suggests that there is a certain degree of overlap in the biological causes of frailty and Multimorbidity, and there may be a bidirectional causal relationship between them. Frailty may make the elderly prone to multiple chronic diseases, but frailty may also stem from the coexistence of multiple diseases [17, 47]. In a systematic review and meta-analysis of 48 observational studies on frailty and Multimorbidity, the possibility of physical frailty due to multiple diseases increased nearly twofold. According to longitudinal studies, Multimorbidity is associated with an increased risk of frailty [48].

Depression in the elderly is also positively correlated with the occurrence of frailty. Previous studies have also found a positive correlation between depression and frailty [49]. Researchers have found that depressive symptoms in frail patients are associated with a higher incidence of frailty (OR = 3.13; 95% CI = 1.37–7.13) [50].

In terms of pathophysiological mechanisms, Interleukin-6 (IL-6) may play a role, as studies show that patients with depression have elevated levels of inflammatory cytokines [51, 52]. High levels of inflammatory cytokines enter the blood-brain barrier and act on skeletal muscles, adversely affecting muscle strength, mass, and cognitive function, leading to the development of frailty [53].

Self-rated health in the elderly is negatively correlated with the occurrence of frailty. Previous studies have also found that frailty is positively correlated with poor health status ($p \leq 0.05$). Elderly people with poor health status have a lower level of education, leading to a significant reduction in self-management ability ($P \leq 0.001$), and a higher degree of frailty ($P \leq 0.001$) [54].

Social events among the elderly are also negatively correlated with the occurrence of frailty. Previous research has also shown that social activities, such as visiting friends, seeking advice, and engaging in intellectual activities, can reduce the decline in Instrumental Activities of Daily Living (IADL) [55]. Participating in meaningful social events contributes to the well-being of the elderly [56]. For the elderly, in order to maintain their mental health, participating in meaningful social activities can increase physical activity and reduce cognitive decline [57].

Our research also indicates that drinking is positively correlated with the occurrence of frailty. This is inconsistent with previous studies, some of which found that moderate drinking can reduce the risk of frailty in elderly men and women, and that consuming 1–7 alcoholic beverages per week is associated with a reduction in the degree of frailty in women ($OR = 0.49$, $p < 0.001$) [58]. Another Mendelian randomization study found no significant causal relationship between alcohol use and the risk of frailty [59]. Other research has found that the risk of frailty is lowest among moderate drinkers who drink 3–4 days per week, possibly showing a U-shaped relationship [60]. Heavy drinkers may be a high-risk group for physical frailty, sick abstainers or those not intending to start drinking are classified as non-drinkers, and healthy drinkers who continue to drink are classified as current drinkers, resulting in a significantly lower risk of frailty among drinkers [61]. Therefore, no definitive conclusions can be drawn about the clear relationship between heavy drinking and frailty. This is a problem that needs to be further addressed in future research.

This study also has several limitations. First, the current definition of frailty is not unified, and we use the FI of the well to assess frailty in the elderly, which may lead to bias in the assessment results. A standard definition of frailty is needed to make the data methods effective and comparable. In the case of cross-sectional studies, we cannot establish a causal relationship between the factors included and the results (frailty). Secondly, the

scope of this study's data survey was during the COVID-19 pandemic in 2020, and new data after the pandemic has not yet been released. Subsequently, with the release of data after COVID-19, a new longitudinal data model can be constructed. Finally, despite taking some measures to ensure good model performance, once the model is established, it still needs to be validated using an independent external queue. Additionally, while we systematically optimized key hyperparameters such as `n_estimators`, `max_depth`, and `learning_rate`, other hyperparameters—such as `num_leaves`, `min_child_samples`, and subsampling techniques—were not included in this study. Future research could explore their impact to further refine the predictive capacity of LightGBM. Despite these limitations, this study shows that the constructed model can be used for the prediction and further intervention of frailty in the elderly.

Conclusion

This study constructed a prediction model for the occurrence of frailty in the elderly in our country using six ML algorithms, and the prediction model based on the LightGBM algorithm has superior comprehensive performance compared to other models. Therefore, the research results can provide a theoretical reference for accurately identifying high-risk groups from multiple dimensions and implementing appropriate early interventions.

Abbreviations

CHARLS	China Health and Retirement Longitudinal Study
AUC	Area under the curve
ROC	Receiver operating characteristic
SHAP	SHapley Additive exPlanations
SVM	Support Vector Machine
PPV	Positive predictive value
ML	Machine learning
NPV	Negative predictive value
KNN	K-Nearest Neighbors
RF	Random Forest
DT	Decision Tree
SMOTE	Synthetic Minority Oversampling Technique
DCA	Decision Curve Analysis
Katz-ADL	Katz activities of daily living
Lawton-IADL	Lawton instrumental activities of daily living
CESD10	Center for Epidemiological Studies Depression Scale

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Authors' contributions

SYLQ conceived the study, participated in its design and coordination, and critically revised the manuscript. SYLQ had full access to all the data collection, analysis, and interpretation, and drafted the manuscript. SYLQ contributed to the process of data collection and data analyses as study investigators. All

authors approved the final manuscript. LY are the guarantors. All authors had full access to all the data in the study, and the corresponding authors had final responsibility for the decision to submit for publication. The corresponding author (LY) attests that all listed authors meet authorship criteria and that no others meeting the criteria have been omitted.

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Data availability

The acquisition of the data is a public database, and the link is: <https://charls.charlsdata.com/pages/data/111/zh-cn.html>.

Declarations

Ethics approval and consent to participate

All methods concerning human participants in our study were conducted in accordance with the ethical standards laid out in the 1964 Declaration of Helsinki and its subsequent amendments. The ethical approval for the 2020 CHARLS survey was granted by the Biomedical Ethics Committee of Peking University under approval number IRB00001052-11015.

Consent for publication

During the field survey, each consenting respondent was required to sign two informed consent forms, with one copy retained by the respondent and the other deposited in the CHARLS office. This manuscript does not include any identifying images or other personal or clinical details of participants that compromise anonymity. Therefore, patient consent for publication is not applicable.

Competing interests

The authors declare no competing interests.

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References

- Zhang M, Chen C, Zhou X, Wang X, Wang B, Huan F, et al. The impact of accelerating population aging on service industry development: evidence from China. *PLoS One*. 2024;19(6):e0296623. <https://doi.org/10.1371/journal.pone.0296623>.
- Deng Y, Zhang K, Zhu J, Hu X, Liao R. Healthy aging, early screening, and interventions for frailty in the elderly. *Biosci Trends*. 2023;17(4):252–61. <https://doi.org/10.5582/bst.2023.01204>.
- Yi M, Zhang W, Zhang X, Zhou J, Wang Z. The effectiveness of Otago exercise program in older adults with frailty or pre-frailty: a systematic review and meta-analysis. *Arch Gerontol Geriatr*. 2023;114:105083. <https://doi.org/10.1016/j.archger.2023.105083>.
- Pek K, Chew J, Lim JP, Yew S, Tan CN, Yeo A, et al. Social frailty is independently associated with mood, nutrition, physical performance, and physical activity: insights from a theory-guided approach. *Int J Environ Res Public Health*. 2020;17(12):4239. <https://doi.org/10.3390/ijerph17124239>.
- Dent E, Morley JE, Cruz-Jentoft AJ, Woodhouse L, Rodríguez-Mañas L, Fried LP, et al. Physical frailty: ICFSR international clinical practice guidelines for identification and management. *J Nutr Health Aging*. 2019;23(9):771–87. <https://doi.org/10.1007/s12603-019-1273-z>.
- Collard RM, Boter H, Schoevers RA, Oude Voshaar RC. Prevalence of frailty in community-dwelling older persons: a systematic review. *J Am Geriatr Soc*. 2012;60(8):1487–92. <https://doi.org/10.1111/j.1532-5415.2012.04054.x>.
- Ma L, Tang Z, Zhang L, Sun F, Li Y, Chan P. Prevalence of frailty and associated factors in the community-dwelling population of China. *J Am Geriatr Soc*. 2018;66(3):559–64. <https://doi.org/10.1111/jgs.15214>.
- Kamo T, Takayama K, Ishii H, Suzuki K, Eguchi K, Nishida Y, Ramos GC, Barbosa AT, Mendonça JM. Epub 2017 Jan 19. PMID: 28126636. Carneiro.
- Chang SF, Lin PL. Frail phenotype and mortality prediction: a systematic review and meta-analysis of prospective cohort studies. *Int J Nurs Stud*. 2015;52(8):1362–74. <https://doi.org/10.1016/j.nurstu.2015.04.005>.
- Sun XH, Ma T, Yao S, Chen ZK, Xu WD, Jiang XY, et al. Associations of sleep quality and sleep duration with frailty and pre-frailty in an elderly population: Rugao longevity and ageing study. *BMC Geriatr*. 2020;20(1):9. <https://doi.org/10.1186/s12877-019-1407-5>.
- Sajeev S, Champion S, Maeder A, Gordon S. Machine learning models for identifying pre-frailty in community dwelling older adults. *BMC Geriatr*. 2022;22(1):794. <https://doi.org/10.1186/s12877-022-03475-9>.
- Licher S, Leening MJG, Yilmaz P, Wolters FJ, Heeringa J, Bindels PJE, et al. Development and validation of a dementia risk prediction model in the general population: an analysis of three longitudinal studies. *Am J Psychiatry*. 2019;176(7):543–51. <https://doi.org/10.1176/appi.ajp.2018.18050566>.
- Hajifathalian K, Ueda P, Lu Y, Woodward M, Ahmadvand A, Aguilar-Salinas CA, et al. A novel risk score to predict cardiovascular disease risk in National populations (Globorisk): a pooled analysis of prospective cohorts and health examination surveys. *Lancet Diabetes Endocrinol*. 2015;3(5):339–55. [https://doi.org/10.1016/S2213-8587\(15\)00081-9](https://doi.org/10.1016/S2213-8587(15)00081-9).
- Dengler NF, Madai VI, Unteroberdörster M, Zihni E, Brune SC, Hilbert A, et al. Outcome prediction in aneurysmal subarachnoid hemorrhage: a comparison of machine learning methods and established clinico-radiological scores. *Neurosurg Rev*. 2021;44(5):2837–46. <https://doi.org/10.1007/s10143-020-01453-6>.
- Akkari G, Nikkhoo M, Wang L, Chen CPC, Han DS, Lin YH, et al. Frailty level classification of the community elderly using Microsoft Kinect-based skeleton pose: a machine learning approach. *Sensors (Basel)*. 2021;21(12):4017. <https://doi.org/10.3390/s21124017>.
- Sandrucci S. Frailty. How to assess, prognostic role. *Eur J Surg Oncol*. 2024;50(5):106862. <https://doi.org/10.1016/j.ejso.2023.03.002>. Epub 2023 Mar 5. PMID: 36922252.
- Fried LP, Tangen CM, Walston J, Newman AB, Hirsch C, Gottdiener J, Seeman T, Tracy R, Kop WJ, Burke G, McBurnie MA. Cardiovascular Health Study Collaborative Research Group. Frailty in older adults: evidence for a phenotype. *J Gerontol A Biol Sci Med Sci*. 2001;56(3):M146–56. <https://doi.org/10.1093/gerona/56.3.m146>. PMID: 11253156.
- Mitnitski AB, Mogilner AJ, Rockwood K. Accumulation of deficits as a proxy measure of aging. *Sci World J*. 2001;1:323–36. <https://doi.org/10.1100/tsw.2001.58>.
- Leghissa M, Carrera Á, Iglesias CA. Machine learning approaches for frailty detection, prediction and classification in elderly people: a systematic review. *Int J Med Inform*. 2023;178:105172. <https://doi.org/10.1016/j.ijmedinf.2023.105172>.
- Lebrasseur A, Fortin-Bédard N, Lettre J, Raymond E, Bussi eres EL, Lapi erre N, et al. Impact of the COVID-19 pandemic on older adults: rapid review. *JMIR Aging*. 2021;4(2):e26474. <https://doi.org/10.2196/26474>.
- K onig M, Gollasch M, Komleva Y. Frailty after COVID-19: the wave after? *Aging Med*. 2023;6(3):307–16. <https://doi.org/10.1002/agm2.12258>.
- Maltese G, Corsonello A, Di Rosa M, Soraci L, Vitale C, Corica F, et al. Frailty and COVID-19: a systematic scoping review. *J Clin Med*. 2020;9(7):2106. <https://doi.org/10.3390/jcm9072106>.
- Liu Y, Cui J, Cao L, Stubbendorff A, Zhang S. Association of depression with incident sarcopenia and modified effect from healthy lifestyle: the first longitudinal evidence from the CHARLS. *J Affect Disord*. 2024;344:373–9. <https://doi.org/10.1016/j.jad.2023.10.012>.
- Zhao Y, Hu Y, Smith JP, Strauss J, Yang G. Cohort profile: the China health and retirement longitudinal study (CHARLS). *Int J Epidemiol*. 2014;43(1):61–8. <https://doi.org/10.1093/ije/dys203>.
- Wang Z, Fang Y, Zhang X. Impact of social capital on health behaviors of middle-aged and older adults in China—an analysis based on CHARLS2020 data. *Healthcare*. 2024;12(11):1154. <https://doi.org/10.3390/healthcare12111154>.
- Mitnitski A, Song X, Skoog I, Broe GA, Cox JL, Grunfeld E, et al. Relative fitness and frailty of elderly men and women in developed countries and their relationship with mortality. *J Am Geriatr Soc*. 2005;53(12):2184–9. <https://doi.org/10.1111/j.1532-5415.2005.00506.x>.
- Rockwood K, Mitnitski A. Frailty in relation to the accumulation of deficits. *J Gerontol Biol Sci Med Sci*. 2007;62(7):722–7. <https://doi.org/10.1093/gerona/62.7.722>.
- Yu Q, Li Z, Yang C, Zhang L, Xing M, Li W. Predicting functional dependency using machine learning among a middle-aged and older Chinese population. *Arch Gerontol Geriatr*. 2023;115:105124. <https://doi.org/10.1016/j.archger.2023.105124>.
- Lei X, Sun X, Strauss J, Zhang P, Zhao Y. Depressive symptoms and SES among the mid-aged and elderly in China: evidence from the China health and

- retirement longitudinal study National baseline. *Soc Sci Med*. 2014;120:224–32. <https://doi.org/10.1016/j.socscimed.2014.09.028>.
30. Boey KW. Cross-validation of a short form of the CES-D in Chinese elderly. *Int J Geriatr Psychiatry*. 1999;14(8):608–17. [https://doi.org/10.1002/\(sici\)1099-1166\(199908\)14:8%3C608::aid-gps991%3E3.0.co;2-z](https://doi.org/10.1002/(sici)1099-1166(199908)14:8%3C608::aid-gps991%3E3.0.co;2-z).
31. Crimmins EM, Kim JK, Langa KM, Weir DR. Assessment of cognition using surveys and neuropsychological assessment: the health and retirement study and the Aging, Demographics, and memory study. *J Gerontol B Psychol Sci Soc Sci*. 2011;66(1):i162–71. <https://doi.org/10.1093/geronb/gbr048>.
32. Booth M. Assessment of physical activity: an international perspective. *Res Q Exerc Sport*. 2000;71(Suppl 2):114–20. <https://doi.org/10.1080/02701367.2000.11082794>.
33. Cruz-Ausejo L, Osada J, Rueda-Torres L, Ingunza Lastra NB, Carrasco-Muñoz MA, Vera-Ponce VJ. Physical activity level and factors associated with perceived stress among Peruvian university professors during the COVID-19 pandemic. *Heliyon*. 2023;9(6):e16439. <https://doi.org/10.1016/j.heliyon.2023.e16439>.
34. Nemoto Y, Sato S, Kitabatake Y, Nakamura M, Takeda N, Maruo K, et al. Longitudinal associations of social group engagement with physical activity among Japanese older adults. *Arch Gerontol Geriatr*. 2021;92:104259. <https://doi.org/10.1016/j.archger.2020.104259>.
35. Lear SA, Hu W, Rangarajan S, Gasevic D, Leong D, Iqbal R, et al. The effect of physical activity on mortality and cardiovascular disease in 130 000 people from 17 high-income, middle-income, and low-income countries: the PURE study. *Lancet*. 2017;390(10113):2643–54. [https://doi.org/10.1016/S0140-6736\(17\)31634-3](https://doi.org/10.1016/S0140-6736(17)31634-3).
36. Wei Z, Wang X, Ren L, Liu C, Liu C, Cao M, Feng Y, Gan Y, Li G, Liu X, Liu Y, Yang L, Deng Y. Using machine learning approach to predict depression and anxiety among patients with epilepsy in china: A cross-sectional study. *J Affect Disord*. 2023;336:1–8. Epub 2023 May 18. PMID: 37209912.
37. Leme DEDC, de Oliveira C. Machine learning models to predict future frailty in community-dwelling middle-aged and older adults: the ELSA cohort study. *J Gerontol A Biol Sci Med Sci*. 2023;78(11):2176–84. <https://doi.org/10.1093/gerona/glad127>.
38. Zhao L, Leng Y, Hu Y, Xiao J, Li Q, Liu C, et al. Understanding decision curve analysis in clinical prediction model research. *Postgrad Med J*. 2024;100(1185):512–5. <https://doi.org/10.1093/postmj/qgae027>.
39. Yang X, Qiu H, Wang L, Wang X. Predicting colorectal cancer survival using time-to-event machine learning: retrospective cohort study. *J Med Internet Res*. 2023;25:e44417. <https://doi.org/10.2196/44417>.
40. Cadrin-Tourigny J, Bosman LP, Nozza A, Wang W, Tadros R, Bhonsale A, Bourfiss M, Fortier A, Lie ØH, Saguner AM, Svensson A, Andorin A, Tichnell C, Murray B, Zeppenfeld K, van den Berg MP, Asselbergs FW, Wilde AAM, Krahn AD, Talajic M, Rivard L, Chelko S, Zimmerman SL, Kamel IR, Crosson JE, Judge DP, Yap SC, van der Heijden JF, Tandri H, Jongbloed JDH, Guertin MC, van Tintelen JP, Platonov PG, Duru F, Haugaa KH, Khairy P, Hauer RNW, Calkins H, Te Riele ASJM, James CA. A new prediction model for ventricular arrhythmias in arrhythmogenic right ventricular cardiomyopathy. *Eur Heart J*. 2022;43(32):e1–9. <https://doi.org/10.1093/eurheartj/ehac180>.
41. Hijazi Z, Oldgren J, Lindbäck J, Alexander JH, Connolly SJ, Eikelboom JW, et al. The novel biomarker-based ABC (age, biomarkers, clinical history)-bleeding risk score for patients with atrial fibrillation: a derivation and validation study. *Lancet*. 2016;387(10035):2302–11. [https://doi.org/10.1016/S0140-6736\(16\)00741-8](https://doi.org/10.1016/S0140-6736(16)00741-8).
42. Li X, Zhao Y, Zhang D, Kuang L, Huang H, Chen W, Fu X, Wu Y, Li T, Zhang J, Yuan L, Hu H, Liu Y, Zhang M, Hu F, Sun X, Hu D. Development of an interpretable machine learning model associated with heavy metals' exposure to identify coronary heart disease among US adults via SHAP: findings of the US NHANES from 2003 to 2018. *Chemosphere*. 2023;311(Pt 1):137039. <https://doi.org/10.1016/j.chemosphere.2022.137039>.
43. Zuo D, Yang L, Jin Y, Qi H, Liu Y, Ren L. Machine learning-based models for the prediction of breast cancer recurrence risk. *BMC Med Inform Decis Mak*. 2023;23(1):276. <https://doi.org/10.1186/s12911-023-02377-z>.
44. Yang CC, Chen PH, Yang CH, Dai CY, Luo KH, Chen TH, et al. Physical frailty identification using machine learning to explore the 5-item FRAIL scale, cardiovascular health study index, and study of osteoporotic fractures index. *Front Public Health*. 2024;12:1303958. <https://doi.org/10.3389/fpubh.2024.1303958>.
45. Marengoni A, Angleman S, Melis R, Mangialasche F, Karp A, Garmen A, et al. Aging with multimorbidity: a systematic review of the literature. *Ageing Res Rev*. 2011;10(4):430–9. <https://doi.org/10.1016/j.arr.2011.03.003>.
46. Calderón-Larrañaga A, Vetrano DL, Onder G, Gimeno-Feliu LA, Coscollar-Santaliestra C, Carfi A, Pisciotta MS, Angleman S, Melis RJF, Santoni G, Mangialasche F, Rizzuto D, Welmer AK, Bernabei R, Prados-Torres A, Marengoni A, Fratiglioni L. Assessing and measuring chronic Multimorbidity in the older population: A proposal for its operationalization. *J Gerontol Biol Sci Med Sci*. 2017;72(10):1417–23. <https://doi.org/10.1093/gerona/glw233>. PMID: 28003375; PMCID: PMC5861938.
47. Cesari M, Pérez-Zepeda MU, Marzetti E. Frailty and multimorbidity: different ways of thinking about geriatrics. *J Am Med Dir Assoc*. 2017;18(4):361–4. <https://doi.org/10.1016/j.jamda.2016.12.086>.
48. Vetrano DL, Palmer K, Marengoni A, Marzetti E, Lattanzio F, Roller-Wirnsberger R, et al. Frailty and multimorbidity: a systematic review and meta-analysis. *J Gerontol A Biol Sci Med Sci*. 2019;74(5):659–66. <https://doi.org/10.1093/geron/a/gly110>.
49. Mezuk B, Edwards L, Lohman M, Choi M, Lapane K. Depression and frailty in later life: a synthetic review. *Int J Geriatr Psychiatry*. 2012;27(9):879–92. <https://doi.org/10.1002/gps.2807>.
50. Jürschik P, Nunin C, Botigué T, Escobar MA, Lavedán A, Viladrosa M. Prevalence of frailty and factors associated with frailty in the elderly population of Lleida, Spain: the FRALLE survey. *Arch Gerontol Geriatr*. 2012;55(3):625–31. <https://doi.org/10.1016/j.archger.2012.07.002>.
51. Vaughan L, Corbin AL, Goveas JS. Depression and frailty in later life: a systematic review. *Clin Interv Aging*. 2015;10:1947–58. <https://doi.org/10.2147/CIA.S69632>. PMID: 26719681; PMCID: PMC4687619.
52. Soysal P, Veronese N, Thompson T, Kahl KG, Fernandes BS, Prina AM, Solmi M, Schofield P, Koyanagi A, Tseng PT, Lin PY, Chu CS, Cosco TD, Cesari M, Carvalho AF, Stubbs B. Relationship between depression and frailty in older adults: A systematic review and meta-analysis. *Ageing Res Rev*. 2017;36:78–87. Epub 2017 Mar 31. PMID: 28366616.
53. Wang C, Zhang J, Hu C, Wang Y. Prevalence and risk factors for cognitive frailty in aging hypertensive patients in China. *Brain Sci*. 2021;11(8):1018. <https://doi.org/10.3390/brainsci11081018>.
54. Cramm JM, Twisk J, Nieboer AP. Self-management abilities and frailty are important for healthy aging among community-dwelling older people; a cross-sectional study. *BMC Geriatr*. 2014;14:28. <https://doi.org/10.1186/1471-2318-14-28>.
55. Fujiwara Y, Shinkai S, Kumagai S, Amano H, Yoshida Y, Yoshida H, et al. Longitudinal changes in higher-level functional capacity of an older population living in a Japanese urban community. *Arch Gerontol Geriatr*. 2003;36(2):141–53. [https://doi.org/10.1016/S0167-4943\(02\)00081-X](https://doi.org/10.1016/S0167-4943(02)00081-X).
56. Cho D, Post J, Kim SK. Comparison of passive and active leisure activities and life satisfaction with aging. *Geriatr Gerontol Int*. 2018;18(3):380–6. <https://doi.org/10.1111/ggi.13188>.
57. Wang HX, Xu W, Pei JJ. Leisure activities, cognition and dementia. *Biochim Biophys Acta*. 2012;1822(3):482–91. <https://doi.org/10.1016/j.bbdis.2011.09.02>.
58. Shah M, Paulson D, Nguyen V. Alcohol use and frailty risk among older adults over 12 years: the health and retirement study. *Clin Gerontol*. 2018;41(4):315–25. <https://doi.org/10.1080/07317115.2017.1364681>.
59. Guo X, Tang P, Zhang L, Li R. Tobacco and alcohol consumption and the risk of frailty and falling: a Mendelian randomisation study. *J Epidemiol Community Health*. 2023;77(6):349–54. <https://doi.org/10.1136/jech-2022-219855>.
60. Kojima G, Liljas A, Iliffe S, Jivraj S, Walters K. A systematic review and meta-analysis of prospective associations between alcohol consumption and incident frailty. *Age Ageing*. 2018;47(1):26–34. <https://doi.org/10.1093/ageing/afx086>.
61. Seematter-Bagnoud L, Spagnoli J, Büla C, Santos-Eggimann B. Alcohol Use and Frailty in Community-Dwelling Older Persons Aged 65 to 70 Years. *J Frailty Aging*. 2014;3(1):9–14. <https://doi.org/10.14283/jfa.2014.2>. PMID: 27049819.

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